

Semantic Theory-guided Approach to Research Synthesis - STARS

Workshop Wuhan, July 2026

1

Your experience with literature reviews

- Topics?
- Methodology (Narrative, systematic, scoping,...)?
- Tools used?
- Published?

2

The systematic review

“Systematic reviews differ from traditional narrative reviews by adopting a **replicable, scientific and transparent** process, in other words, a detailed technology, that aims to minimize bias through **exhaustive** literature searches of published and unpublished studies and by providing an **audit trail of the reviewer’s decisions, procedures and conclusions.**”

(Tranfield et al., 2003: 209)

3

Reproducible data analysis and FAIR data

For **reproducibility**:

- Minimize manual analysis steps.
- Use only free/open-source software whenever possible.
- Document each step of the analysis.

FAIR data:

- **F**indable
- **A**ccessible
- **I**nteroperable and
- **R**eusable

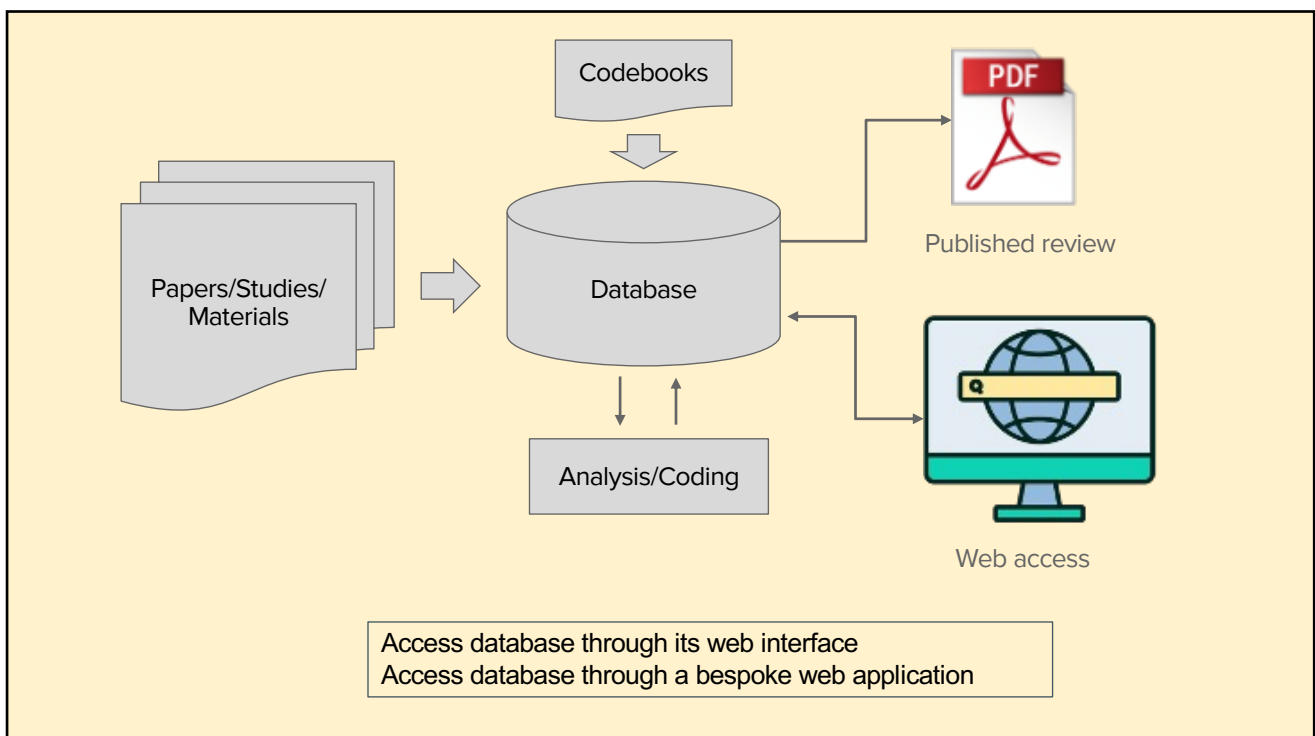
4

Problems with the established literature review

- Outdated at time of publication
- Difficult to keep up-to-date
- Review process difficult to reproduce
- Codebooks not reused
- Unpublished reviews no used
- Intermediate products (partially coded corpus) not (re-)used

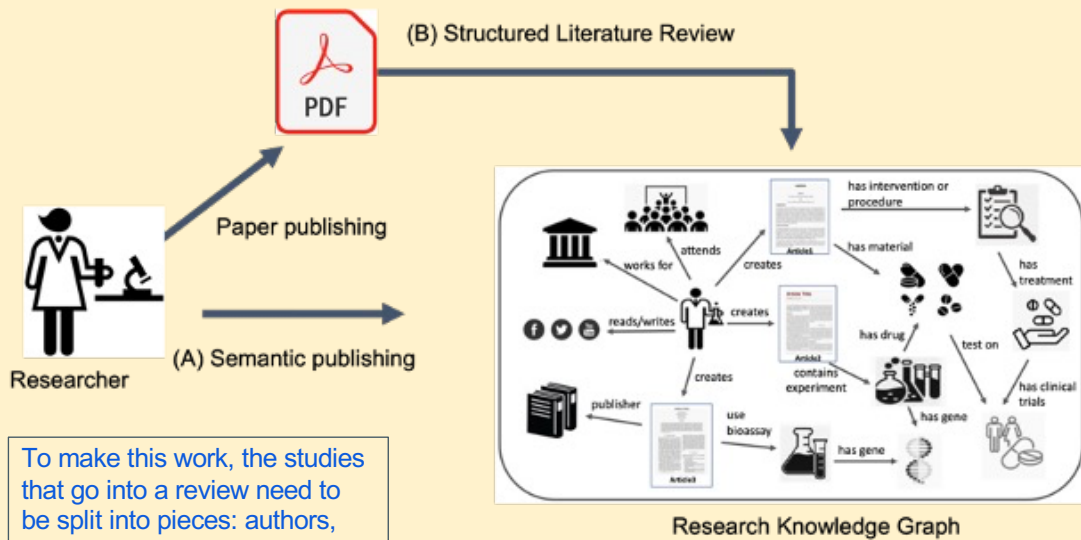


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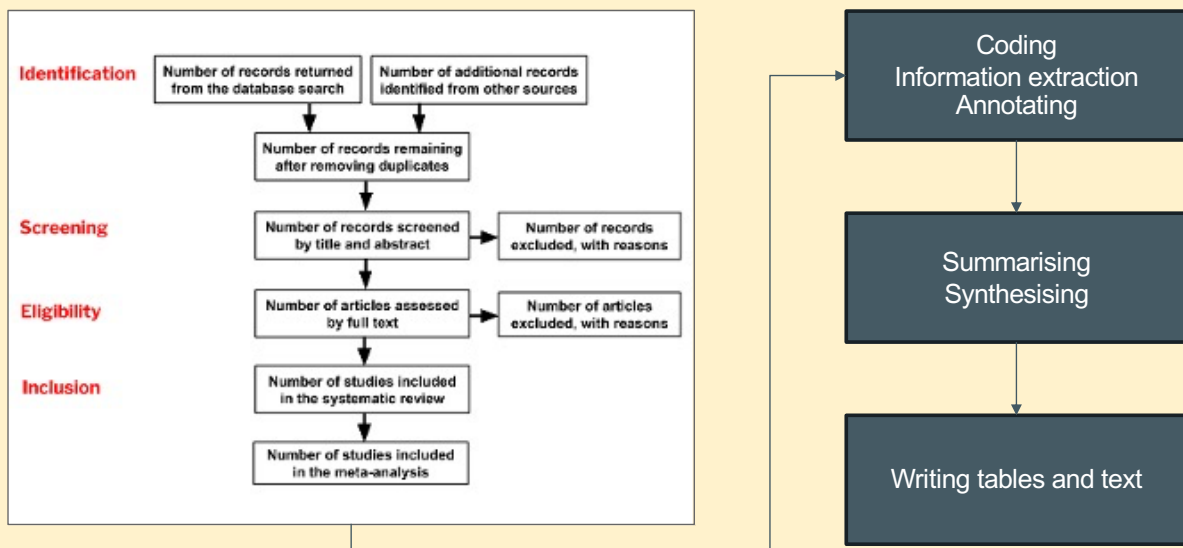
6

Big idea: New ways of publishing



8

STARS starts where PRISMA ends



9

"Coding" a paper =

- Information extraction (e.g., number of participants, age group, methods)
- +
- Classifications (e.g., quality of evidence)
- +
- Text annotations

DE GRUYTER

Nanotechnol Rev 2015; 4(1): 51–69

Nanotechnology education

Cesar Delgado*, Shawn Y. Stevens, Namsoo Shin and Joseph Krajcik

A middle school instructional unit for size and scale contextualized in nanotechnology

Abstract: Size and scale is a "big idea" in nanoscale science and engineering and is poorly understood by secondary students. This paper describes the design process, implementation, and evaluation of a 12-h instructional unit for size and scale, in a summer science camp for middle school students from a low SES public school district. Instructional activities were designed following a construct-centered design approach and included the use of microscopes, custom-made computer simulations, and 2-D and 3-D scale models. The unit followed a project-based instructional approach and was contextualized with the driving question, "How can nanotechnology keep me from getting sick?" Pre- and post-intervention interviews revealed that students significantly increased their qualitative and quantitative knowledge of the size of objects including atom, viruses, and cells, with an effect size of 0.8 for an overall metric. The campers closed the gap with private middle school students and on some measures surpassed high school students from the same district. The principle of "broad spectrum" curriculum and instruction – activities that target specific advanced understandings but simultaneously scaffold or support the learning of more fundamental, prerequisite ideas – was inductively generated from an analysis of the learning activities.

Keywords: curriculum; nanoeducation; scale.

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*Corresponding author: Cesar Delgado, Department of Curriculum and Instruction, STEM Education Program, University of Texas, 1912 Speedway Stop D5500, Austin, TX 78712, USA, e-mail: cesar_delgado@austin.utexas.edu
http://orcid.org/10.1000-0001-9571-1806
Shawn Y. Stevens and Namsoo Shin: Educational Studies, University of Michigan, Ann Arbor, MI, USA
Joseph Krajcik: Science Education, Michigan State University, East Lansing, MI, USA

1 Introduction

Despite repeated calls to incorporate nanoscale science and engineering (NSE) into the pre-college curriculum, current standards hardly mention nanotechnology or nanoscience [1, 2]. In the current educational climate that stresses accountability and high stakes testing keyed to standards, the absence of NSE in the standards may lead to its continued absence in the curriculum. However, several big ideas identified as fundamental in NSE learning [3] are prominent in the new science education standards and are, thus, promising vehicles for introducing NSE topics and prerequisite knowledge at the pre-college level. Among these big ideas is "size and scale," which overlaps with the standards' emphasis on the crosscutting concept of "size, proportion, and quantity." As a crosscutting concept, size and scale is not only a NSE big idea unto itself, but is also important for developing an understanding of learning goals in other NSE big ideas (e.g., Size-Dependent Properties, Forces and Interactions), making its inclusion in pre-college science classes particularly important.

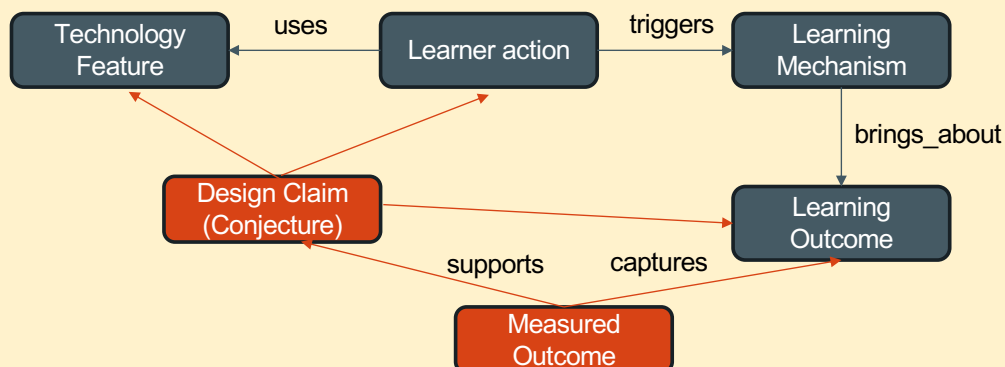
Research shows that US learners have a weak grasp on the size and scale of objects included in the science curriculum, such as atoms, cells, the Sun, or the Earth, particularly at the nanoscale. For instance, the ability of people from fifth grade to experts to name objects of a specific size range falls "precipitously" at sizes lower than a millimeter [4], and 57% of people of various ages were unable to accurately order millimeter, micrometer, and nanometer by size [5]. Thus, there is a need for improved curriculum and instruction for size and scale.

While there are some published instructional activities for size and scale [6–8], few have been evaluated for their effectiveness. One exception is Jones and colleagues' [9] use of the classic Eames film *Powers of Ten* [10] in a lesson with middle school students, where the mean score of students' sorting of 12 objects or distances from diameter of proton to Earth-Sun distance showed a statistically significant improvement of 60.09–62.59, and their ability to match the size in meters using scientific notation

10

Coding an edtech study (a "paper")

- For edtech studies, the key unit to code/model/represent is the learner activity/action (Why?)



11

Coding an edtech study (a “paper”)

Concept	Question
Learning topics and goals	What is the student supposed to learn?
Technology feature	What representation or tool is used?
Activity/Action	What is the learner task? What actions are taken by the learner?
Activity/Action outcome	What are they? (These are available for learning)
Learning outcome	What is learned from the activity (or even the action)
Claim	What is supported by evidence?

12

Copding a paper

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Author	Year	Pedagogy	Features	Actions	Outcomes
Delgado	2015	PBL	Simulation, Microscope	Observe, Measure	Size understanding

13

3.1.3 Lessons

The activities came from various sources. One activity was inspired by Tretter's 50,000:1 scale activity "Conceptualizing Nanoscale" [8], but used a million-to-one scale; at this magnification, an atom would just be visible, students would be the size of bacteria, and a football field would represent the thickness of a hair, thus, spanning the sub-microscopic world from barely visible down to 0.1 nm. Other activities were modified from standard classroom activities. Students visualized and sketched their own cheek cells under an optical microscope, but the slide also had a human hair on it, so they could estimate the relative size of the cheek cell and then calculate its diameter from the known thickness of the hair (see Figure 3).

Students also collaboratively swabbed surfaces for bacteria and incubated them on growth medium in Petri dishes, to test for the presence of bacteria on different parts of the building. Other activities were original, for instance, modeling the effect of surface roughness on how easy it is to clean a surface of bacteria using different grades of sandpaper to represent surfaces and salt to represent bacteria [40]. We designed a computer visualization [41] that shows successive magnifications of objects and

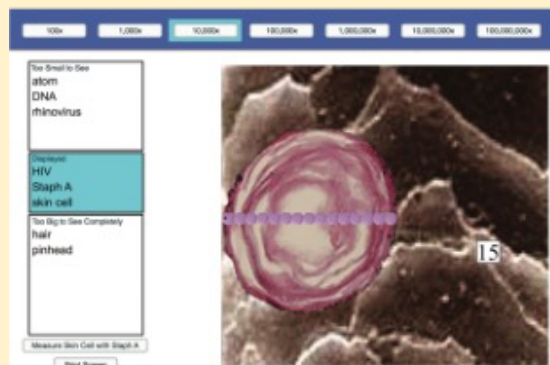


Figure 2: Screenshot of the "size and scale" computer visualization. The counter shows that 15 *S. aureus* bacteria fit across a skin cell.

Appendix 1: Unpacking.

Big idea: One-dimensional Size and Scale
Grade level: middle school (6th–8th grade)

Critical concepts	Prerequisite knowledge	Alternative ideas and findings from previous studies (Citation)	Phenomena (link to a nano product)	Claim	Evidence	Task
1. There are unseen worlds that are too small to be seen with the unaided eye.	Students should know some examples of submicroscopic objects, e.g., microbes (typically covered in elementary), bacteria (middle school), atom and molecules (high school).	15% [20] and 40% [21] of students mentioned a small visible object when asked for the smallest thing which they could think of.	Bacteria and viruses can make us sick (link to nanotile).	Students can compare and contrast the eye and the optical microscope as instruments used to visualize objects.	Students identify the greater resolution of the microscope as an advantage, and the limited field of view as a disadvantage, relative to the eye.	Students observe grains of sand and salt, and a mm ruler under the microscope and with the naked eye, and record their observations. They compare and contrast advantages of each tool.

14

Activity coding exercise

- Students used microscopes to observe and sketch cells and bacteria.
- They estimated sizes using reference objects and compared different scales.
- Later, they used computer simulations to explore ordering of objects across scales.



https://github.com/prei007/wuhan_workshops/tree/main/litreview

Download: coding_exercises.xlsx

15

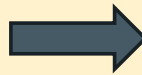
Multiple cell values:

Activity	Feature	Actions	Outcomes
Microscope observation & sketching	Microscope	Observe, Measure, Compare, Estimate	Relative size, Absolute size
Simulation exploration	Simulation	Compare, Order, Measure	Ordering by size, Absolute size
Scale modelling	Scale model	Construct, Compare, Estimate	Size & scale understanding

Sequencing: A1 -> A2 -> A3

Add a column?	
Activity	Next Activity
A2	A3
A3	A4

Where do claims go?



The 3-column (3C) format is more flexible. (see sheet #2)

16

Coding findings and claims

Design claims: Strategy/Feature <x> improves learning (motivation, etc.)

Empirical claims: "This study finds..."

Delgado et al. (2015) found that:

- Students significantly improved their understanding of size and scale (effect size ~0.79).
- However, estimating relative size remained difficult and resistant to change.

What is missing?

17

Delgado et al. (2015) found that:

- Students significantly improved their understanding of size and scale (effect size ~0.79).
- However, estimating relative size remained difficult and resistant to change.

Missing:

- Which activities contributed?
- Dosage?
- Which outcomes exactly?
- By which mechanisms?
- How related to other claims made elsewhere?

Coding claims and findings -> Sheet #3

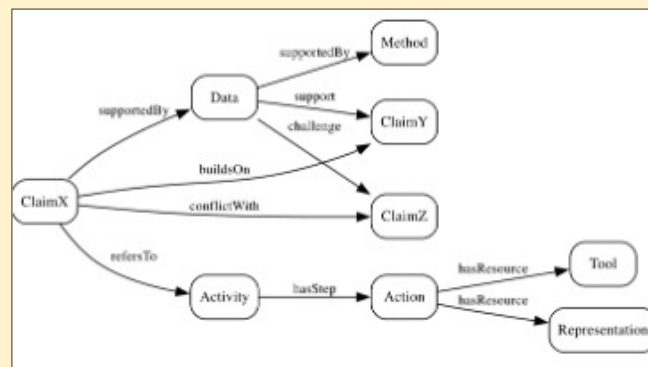
18

Delgado et al. (2015) found that:

- Students significantly improved their understanding of size and scale (effect size ~0.79).
- However, estimating relative size remained difficult and resistant to change.

Missing:

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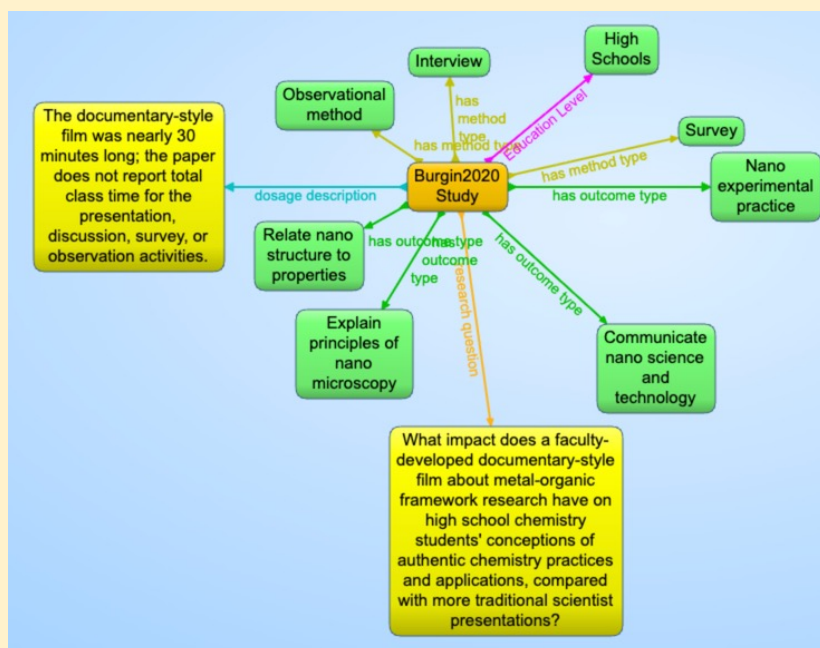
19

Let's put triples into a database

s	p	o
res:Burgin2020_Publication	rdf:type	dcterms:BibliographicResource
res:Burgin2020_Publication	rdfs:label	"Burgin et al. (2020)"
res:Burgin2020_Publication	dcterms:title	"Using a Faculty-Developed Documentary-Style Film to Communicate Authentic Chemistry Research to a High School Audience"
res:Burgin2020_Publication	dcterms:identifier	"10.1021/acs.jchemed.0c00376"
study:Burgin2020_Study	edp:researchQuestion	"What impact does a faculty-developed documentary-style film about metal-organic framework research have on high school chemistry students' conceptions of authentic chemistry practices and applications, compared with more traditional scientist presentations?"
study:Burgin2020_Study	edp:hasMethodType	meth:SurveyMethod
study:Burgin2020_Study	edp:hasMethodType	meth:ObservationMethod
study:Burgin2020_Study	edp:hasMethodType	meth:InterviewMethod
study:Burgin2020_Study	edp:sampleSize_Treatment	"122"
study:Burgin2020_Study	dcterms:educationLevel	edlev:HighSchools

20

It's all connected



21

Aggregating across studies

You might wonder: "How are claims about learning environments and their technology features supported across all studies in the database?"



Translate the question into a query that follows the relevant relations (predicates)



envLabel	featureLabelShown	supportTarget	claim	claimLabel	data	dataLabel
"Lan2021 Digital Twin Technology"	"Design features: Performance and grading system"	"Feature"	Lan2021_System_Clar	"Design claim: Lan2021"	Lan2021_Data	"Observation and interpretation of data from multiple sources"
"Lan2021 Digital Twin Technology"	"Adaptive guidance"	"Feature"	Lan2021_System_Clar	"Design claim: Lan2021"	Lan2021_Data	"Observation and interpretation of data from multiple sources"
"Lan2021 Digital Twin Technology"	"Adaptive guidance"	"Feature"	Lan2021_System_Clar	"Design claim: Lan2021"	Lan2021_Data	"Observation and interpretation of data from multiple sources"
"Lan2021 Digital Twin Technology"	"Design features: Performance and grading system"	"Feature"	Lan2021_System_Clar	"Design claim: Lan2021"	Lan2021_Data	"Observation and interpretation of data from multiple sources"

```
SELECT DISTINCT
  ?envLabel
  ?featureLabelShown
  ?supportTarget
  ?claim ?claimLabel
  ?data ?dataLabel
WHERE {
  ?env a edp:LearningEnvironment ;
    edp:hasFeature ?feature .
  {
    ?feature a edp:Feature ;
      edp:hasSupportingClaim ?claim .
    BIND("Feature" AS ?supportTarget)
  }
  UNION
  {
    ?env edp:hasSupportingClaim ?claim .
    BIND("LearningEnvironment" AS ?supportTarget)
  }
  # Support: Data → Claim
  ?data a mp:Data ;
    mp:supports ?claim .
}
```

23

Did we make progress?

For **reproducibility**:

- Minimize manual analysis steps.
- Use only free/open-source software whenever possible.
- Document each step of the analysis.

FAIR data:

- Findable
- Accessible
- Interoperable and
- Reusable

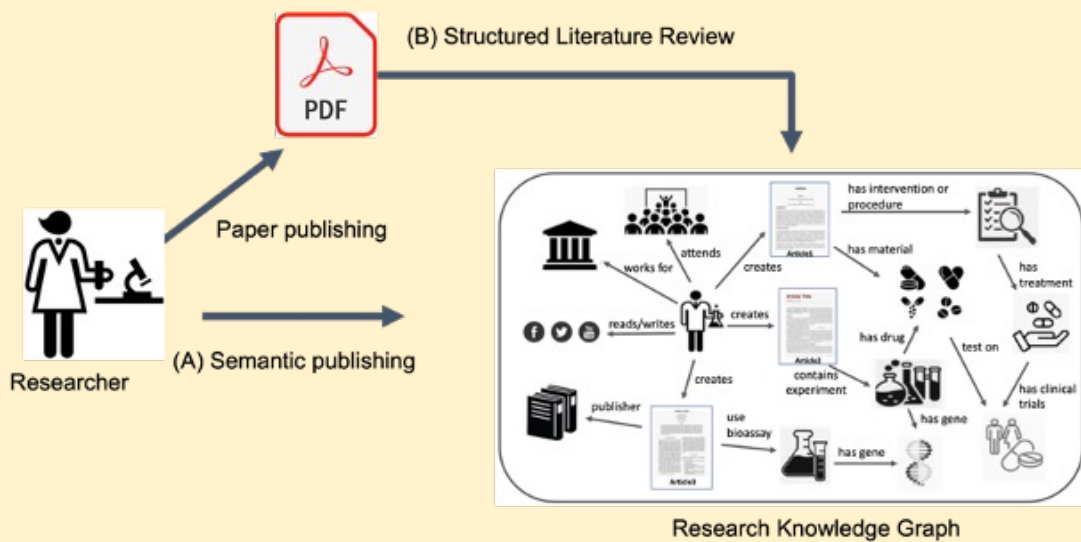
For exploring the power of RDF, I recommend the free version of graphdb (<https://graphdb.ontotext.com/>)

24

Session 2

25

Big idea: publishing knowledge, not only text



26

The power of 3C (three columns)

Subject (resource)	Predicate (relation)	Object (resource)
Action1(A1)	usesFeature	Microscope(F2)
A1	hasActionType	Observe
A1	hasOutcomeType	ScaleAndSizeUnderstanding
Observation	precedes	Simulation
A1	precedes	A2
Data1	effectsize	4.5
Claim1	supportedBy	Data1
Claim1	refersTo	F2
Claim1	authoredBy	PaperReviewer
StudyQuality-Stmt1	hasValue	high
StudyQuality-Stmt1	estimatedBy	PaperReviewer

Relations between different entities

Sequence

Simple facts

Claims, hypotheses

Provenance

Appraisals

27

Up next

- From 3C tables to RDF (Resource Description Framework) graphs
- From list of codes to taxonomies
- Thematic coding

28

From 3C tables to RDF triple statements

Subject (resource)	Predicate (relation)	Object (resource)
Bob	Is a	Person
Bob	Is a friend of	Alice
...	...	...



Triples as plain text statements.

EXAMPLE 1: Sample triples (informal)

```
<Bob> <is a> <person>.
<Bob> <is a friend of> <Alice>.
<Bob> <is born on> <the 4th of July 1990>.
<Bob> <is interested in> <the Mona Lisa>.
<the Mona Lisa> <was created by> <Leonardo da Vinci>.
<the video 'La Joconde à Washington'> <is about> <the Mona Lisa>
```

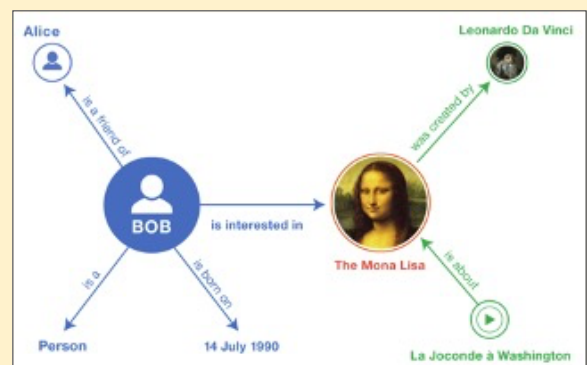
29

From RDF triple statements to graphs



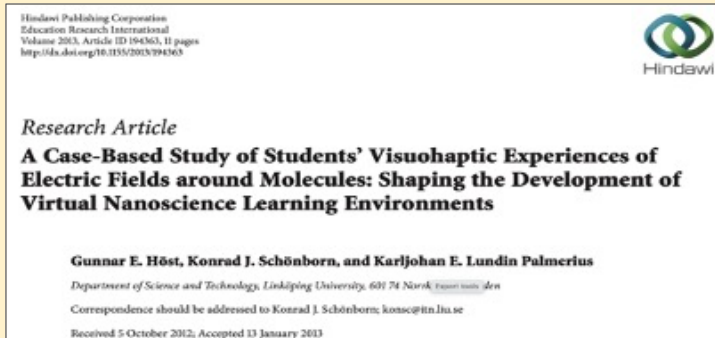
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<the Mona Lisa> <was created by> <Leonardo da Vinci>.
<the video 'La Joconde à Washington'> <is about> <the Mona Lisa>
```



30

Instead of the Mona Lisa



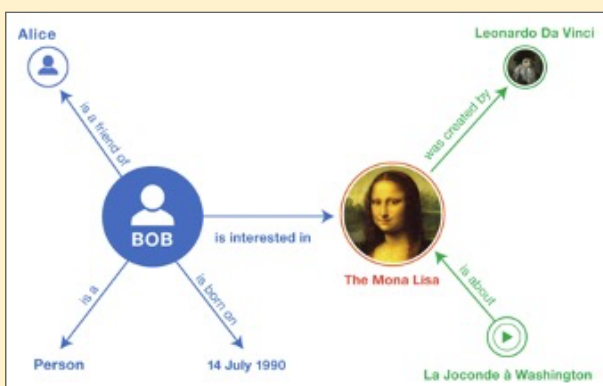
Let's use [Mermaid.live](https://mermaid.live) to formulate some statements.

```

flowchart LR
  Hoest2013 -->|has_author| GH["Gunnar Hoest"]
  Hoest2013 -->|has_title| Title1["Learning from Laboratory Work"]
  Hoest2013 -->|published_in| JRST
  
```

31

Three kinds of relations



Subject — Predicate —> Object
 <Bob> <is interested in> <The Mona Lisa>

Subject — Predicate —> Data
 <Bob> <is born on> <14 July 1990>

Subject — Predicate —> Class
 <Bob> <is a> <Person>

32

Clarifying intended meaning: Namespaces

- They look like this: <https://schema.org/Event>.
- Why are they needed?
 - Because names and concepts are not identical.
- Think of two people with the same name: "John Smith"
 - Same name, different people
- Or two phone numbers in different countries
- We need a country code or a postal address
- Computers need unique identifiers ("postal addresses")
- Example: "Author" in a library database and in a journal database
 - <http://uni-sydney-library.edu.au/Author>
 - <http://springer-nature.com/Author>

33

Namespaces: Why prefixes?

```
http://purl.org/dc/terms/title
http://purl.org/dc/terms/creator
http://purl.org/dc/terms/date
```



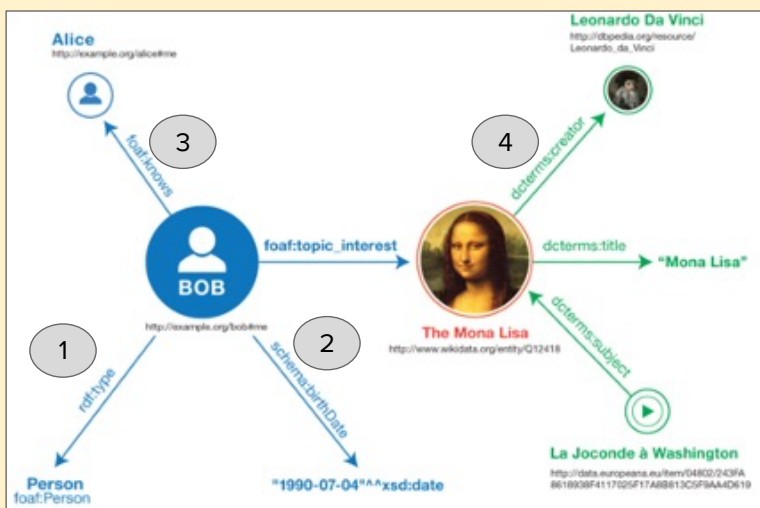
```
@prefix dct: <http://purl.org/dc/terms/> .
```



```
dct:title
dct:creator
dct:date
```

34

RDF vocabularies: Giving resources meaning



- (1) **rdf/rdfs** is about types of resources.
- (2) **schema** is about resources typically mentioned on web pages.
- (3) **foaf** is about people and their relations.
- (4) **dcterms** is about media artefacts.

35

Giving resources meaning

Event		
A Schema.org Type		
Thing > Event		
An event happening at a certain time and location, such as a concert, lecture, or festival. Ticketing information may be added via the offers property. Repeated events may be structured as separate Event objects. [more...]		
Property	Expected Type	Description
Properties from Event		
about	Thing	The subject matter of an object. Inverse property: subjectOf .
actor	PerformingGroup or Person	An actor (individual or a group), e.g. in TV, radio, movie, video games etc., or in an event. Actors can be associated with individual items or with a series, episode, clip. Supersedes actors .

Example: The event schema (<https://schema.org/Event>).

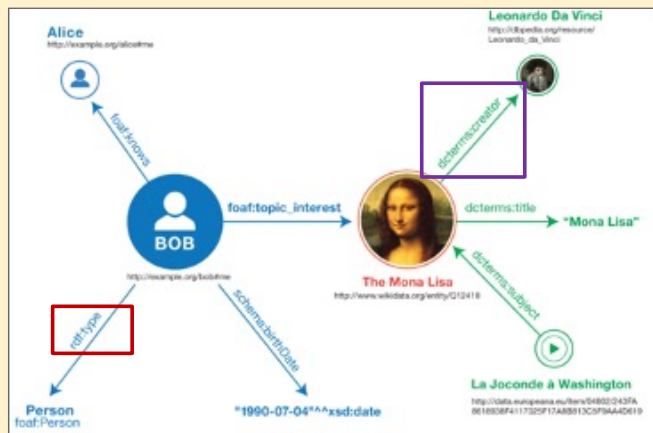
Important: You can create predicates (and classes) as you see fit yourself and describe their intended meaning using RDF Schema (RDFS).

But you should only do so if no suitable standardised predicate exists.

The same holds for classes (such as Event, Person, Research method....)

36

RDF vocabularies



EXAMPLE 6: N-Triples

```

61 <http://example.org/bob#me> <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://xmlns.com/foaf/0.1/Person> .
62 <http://example.org/bob#me> <http://xmlns.com/foaf/0.1/knows> <http://example.org/alice#me> .
63 <http://example.org/bob#me> <http://schema.org/birthDate> "1990-07-04"^^<http://www.w3.org/2001/XMLSchema#date> .
64 <http://example.org/bob#me> <http://xmlns.com/foaf/0.1/topic_interest> <http://www.wikidata.org/entity/Q12418> .
65 <http://www.wikidata.org/entity/Q12418> <http://purl.org/dc/terms/title> "Mona Lisa" .
66 <http://www.wikidata.org/entity/Q12418> <http://purl.org/dc/terms/creator> <http://dbpedia.org/resource/Leonardo_da_Vinci> .
67 <http://data.europeana.eu/item/84882/243F48818938F4117925F17A88813C5F8A4D619> <http://purl.org/dc/terms/subject> <http://www.wikidata.org/entity/Q12418> .

```

37

Two (of many) ways to write spo statements

In a column
format:

subject	predicate	object
Barnett2011	science	Ecology
Barnett2011	educationLevel	HighSchool
...	...	...

In a sentence format ("Turtle notation")

```

:Barnett2011 a dc:BibliographicResource, bibo:Article ;
  litrev:science sci:Ecology ;
  litrev:learningOutcome lo:MultiDomainOutcomes ;
  litrev:educationLevel edlevel:HighSchool .

```

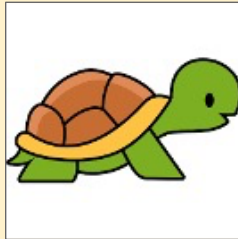
End of statement
group

38

Turtle for readability

EXAMPLE 6: N-Triples

```
01 <http://example.org/bob#me> <http://www.w3.org/1999/02/22-rdf-syntax-ns#type> <http://xmlns.com/foaf/0.1/Person> .
02 <http://example.org/bob#me> <http://xmlns.com/foaf/0.1/knows> <http://example.org/alice#me> .
03 <http://example.org/bob#me> <http://schema.org/birthDate> "1990-07-04"^^<http://www.w3.org/2001/XMLSchema#date> .
04 <http://example.org/bob#me> <http://xmlns.com/foaf/0.1/topic_interest> <http://www.wikidata.org/entity/Q12418> .
05 <http://www.wikidata.org/entity/Q12418> <http://purl.org/dc/terms/title> "Mona Lisa" .
06 <http://www.wikidata.org/entity/Q12418> <http://purl.org/dc/terms/creator> <http://dbpedia.org/resource/Leonardo_da_Vinci> .
07 <http://data.europeana.eu/item/04802/243FA8618938F4117025F17A8B813C5F9AA4D619> <http://purl.org/dc/terms/subject> <http://www.wikidata.org/entity/Q12418> .
```



EXAMPLE 7: Turtle

```
01 BASE <http://example.org/>
02 PREFIX foaf: <http://xmlns.com/foaf/0.1/>
03 PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
04 PREFIX schema: <http://schema.org/>
05 PREFIX dcterms: <http://purl.org/dc/terms/>
06 PREFIX wd: <http://www.wikidata.org/entity/>
07
08 <bob#me>
09   a foaf:Person ;
10   foaf:knows <alice#me> ;
11   schema:birthDate "1990-07-04"^^xsd:date ;
12   foaf:topic_interest wd:Q12418 .
13
14 wd:Q12418
15   dcterms:title "Mona Lisa" ;
16   dcterms:creator <http://dbpedia.org/resource/Leonardo_da_Vinci> .
17
18 <http://data.europeana.eu/item/04802/243FA8618938F4117025F17A8B813C5F9AA4D619>
19   dcterms:subject wd:Q12418 .
```

39

Let's summaries

- There are two kinds of **objects**: classes and individuals/instances
- **Classes** are sets, collections, types of objects, kinds of things
- **Individuals** (or instances) are what a class groups together
- Properties come in two types. An **object property** is a relationship between two things. **Datatype properties** are attributes of one thing
- In graph representation diagrams, classes & individuals are called **nodes** while properties are called **edges**
- In **RDF triples**, classes and individuals are the **subjects** or **objects**, while properties correspond to **predicates**. A set of RDF triples is called an **RDF graph**

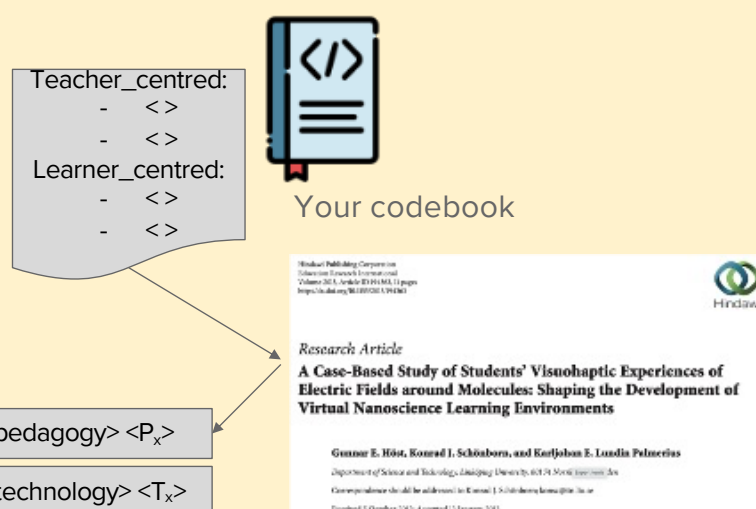
40

RDF for coding in literature reviews

- What have you been coding for?
- How did you come up with your codes?
- How are the codes organised?
 - A flat term list
 - A hierarchical taxonomy
 - (...)
- (How) Did you document the coding process?

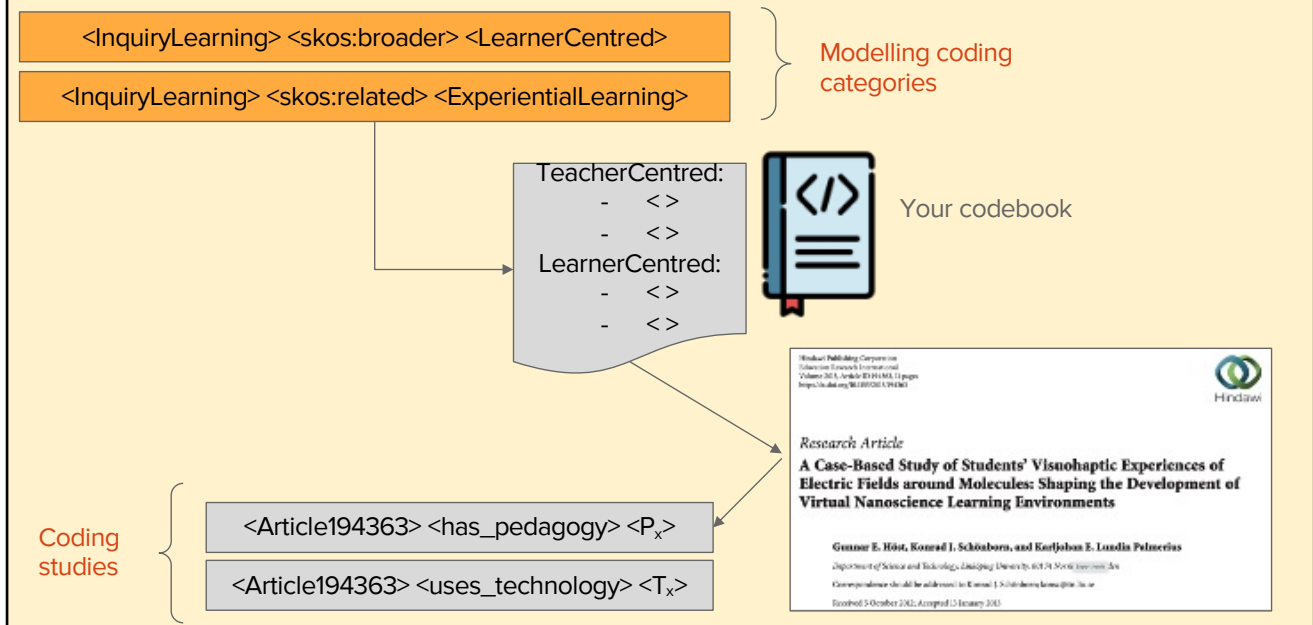
41

An important step: RDF for modelling codebooks



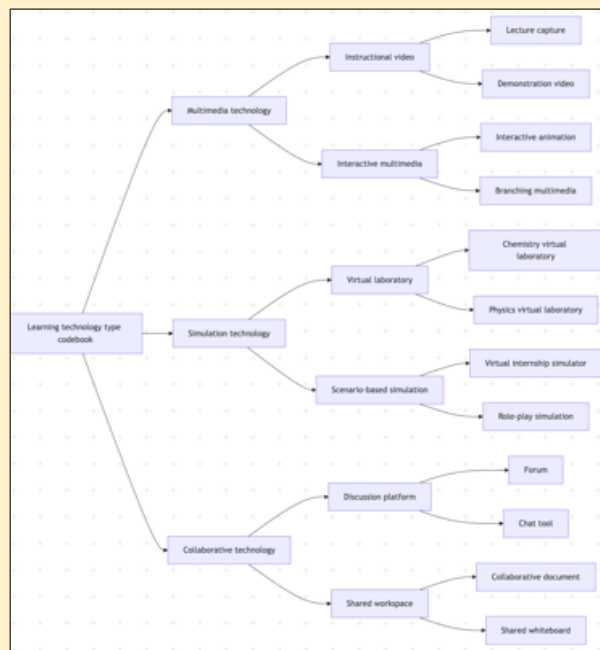
42

An important step: RDF for modelling codebooks



43

Example: A taxonomy for learning technologies



44

```

@prefix skos: <http://www.w3.org/2004/02/skos/core#> .
@prefix lt:   <https://example.org/codebook/learning-technology/> .

lt:TechnologyTypeScheme
  a skos:ConceptScheme ;
  skos:prefLabel "Learning technology type codebook"@en ;
  skos:definition "A three-level coding scheme for classifying
    learning technologies in literature reviews and content
    analysis."@en .

lt:MultimediaTechnology
  a skos:Concept ;
  skos:inScheme lt:TechnologyTypeScheme ;
  skos:prefLabel "Multimedia technology"@en ;
  skos:definition "Learning technologies that combine multiple media
    forms such as text, image, audio, animation, or video."@en ;
  skos:narrower lt:InstructionalVideo ,
    lt:InteractiveMultimedia .

```

45

Using the concept hierarchy in queries

```

PREFIX skos: <http://www.w3.org/2004/02/skos/core#>
PREFIX lt:   <https://example.org/codebook/learning-technology/>

SELECT ?higherConcept ?higherLabel
WHERE {
  lt:VirtualInternship skos:broader+ ?higherConcept .
}

```



Scenario-based simulation
Simulation technology

46

SKOS summary

Concepts	Labels & Notation	Documentation	Semantic Relations	Mapping Properties	Collections
Concept	prefLabel	note	broader	broadMatch	Collection
ConceptScheme	altLabel	changeNote	narrower	narrowMatch	orderedCollection
inScheme	hiddenLabel	definition	related	relatedMatch	member
hasTopConcept	notation	editorialNote	broaderTransitive	closeMatch	memberList
topConceptOf		example	narrowerTransitive	exactMatch	
		historyNote	semanticRelation	mappingRelation	
		scopeNote			

https://en.wikipedia.org/wiki/Simple_Knowledge_Organization_System

47

Exercise

Write a SKOS definition for ``InstructionalVideo``, with

- Definition:
"Video-based learning resources designed to explain, demonstrate, or contextualise concepts and procedures.";
- broader category ``MultimediaTechnology``;
- narrower categories ``LectureCapture`` and ``DemonstrationVideo``.

48

Solution

```
PREFIX skos: <http://www.w3.org/2004/02/skos/core#>
PREFIX lt:   <https://example.org/codebook/learning-technology/>

lt:InstructionalVideo
  a skos:Concept ;
  skos:broader lt:MultimediaTechnology ;
  skos:prefLabel "Instructional video"@en ;
  skos:definition "Video-based learning resources designed to
explain, demonstrate, or contextualise concepts and
procedures."@en ;
  skos:narrower lt:LectureCapture ,
               lt:DemonstrationVideo .
```

https://github.com/prei007/wuhan_workshops/blob/main/litreview/learning_tech_taxonomy

49

How can thematic coding benefit from semantic taxonomies modelled in SKOS?

- We see the relations to other concepts.
- It helps to understand how other use specific concepts.
- Help search engines and LLMs to find what you mean.

50

What does it look like in a fuller example?

- Input: SKOS-coded study
- Storage: Database optimized for RDF (and, hence, SKOS)
- Analysis: Creating tables from codes
- Output:
 - The database on the web.
 - The review data as a website.
 - The review data accessed through a chatbot.

51

What next?

- So far, we have looked only at extracting information from an individual study manually
- LLMs can be used to extract information and model them in RDF
- Claims need more specific attention: From single-study claims to claim networks

52

Claim networks

53

Clark et al. *Journal of Biomedical Semantics* 2014, 5:28
<http://www.jbiomedsem.com/content/5/1/28>



JOURNAL OF
BIOMEDICAL SEMANTICS

RESEARCH

Open Access

Micropublications: a semantic model for claims, evidence, arguments and annotations in biomedical communications

Tim Clark^{1,2,3*}, Paolo N. Ciccarese^{1,2} and Carole A. Goble³

54

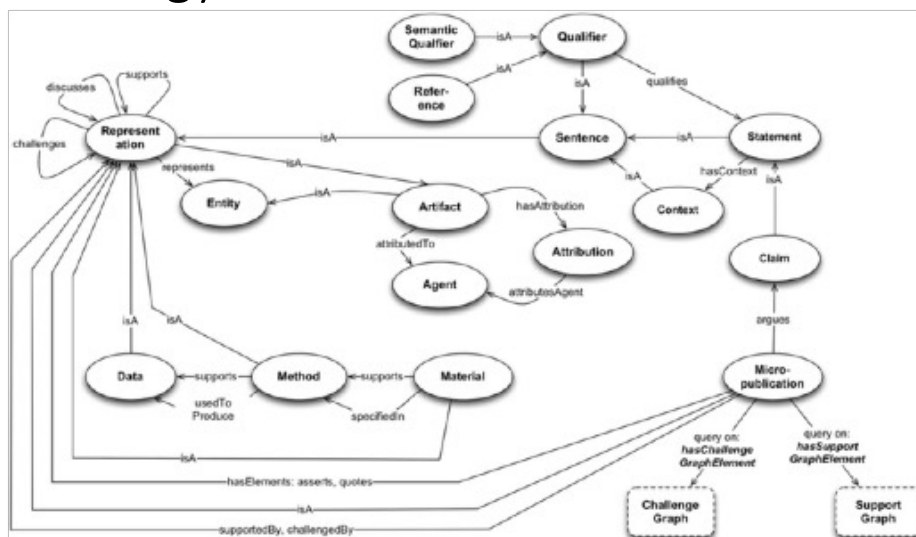
Modelling argumentation relations in research publications

This model, the Micropublications Ontology, is inspired by the Toulmin model of argumentation in order to organize citable claims from scientific papers into networks of agreement and disagreement.

- A minimal **Micropublication** is a **Claim** with **Attribution**.
- Warrants—**Data**, **Materials** and **Methods**—are explicitly represented.
- The **Data** collected is considered meaningful due to a subfield's approval of the **Materials** and **Methods**. This approval is the backing of claims made in everyday science .
- With this model, truth-bearing **Statements** (possibly with **Qualifiers**) are explicitly connected to the **Data**, **Material**, and **Methods**.
- **Attribution** also has a place (for instance justifying use of a method by citing it in the literature).
- A sub-argument can be represented as a **Micropublication** for a related **Claim**, however far back we wish to extend the network.

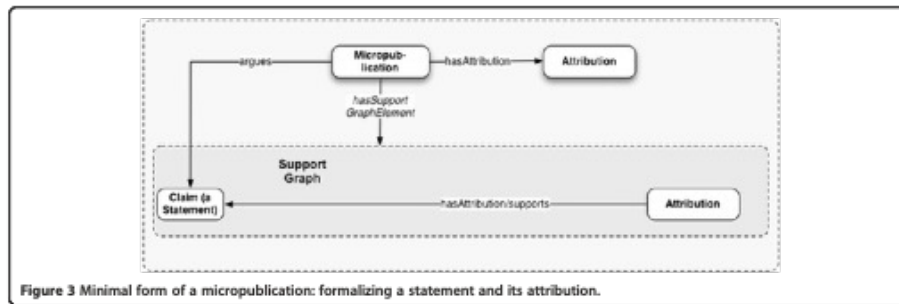
55

mp ontology



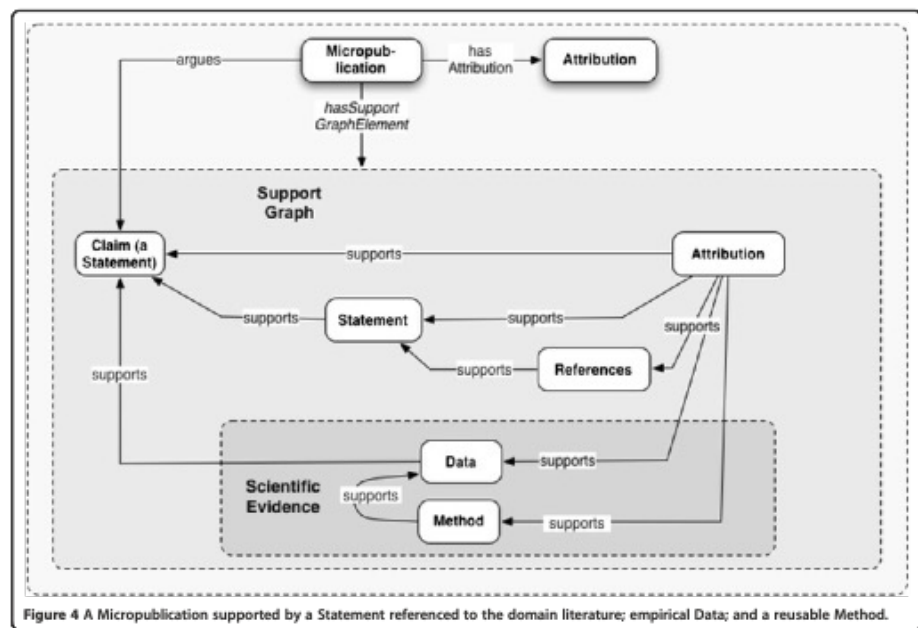
56

Minimal micropublication



57

A more
fully
developed
MP



58

Example study

existing scientific explanations for the given subject (Adair, 2013; Gunstone, 1984; Halloun & Hestenes, 1985). The concept of circular motion (CM) is a phenomenon experienced by everyone daily, thereby creating room for alternative conceptions on the topic before learners are taught the correct conceptual understanding (Canlas, 2016). The current study sought to investigate ways to use a learner-centered approach in the form of an online inquiry-based learning (IBL) environment for teaching CM to first-year undergraduate students in an introductory physics course.

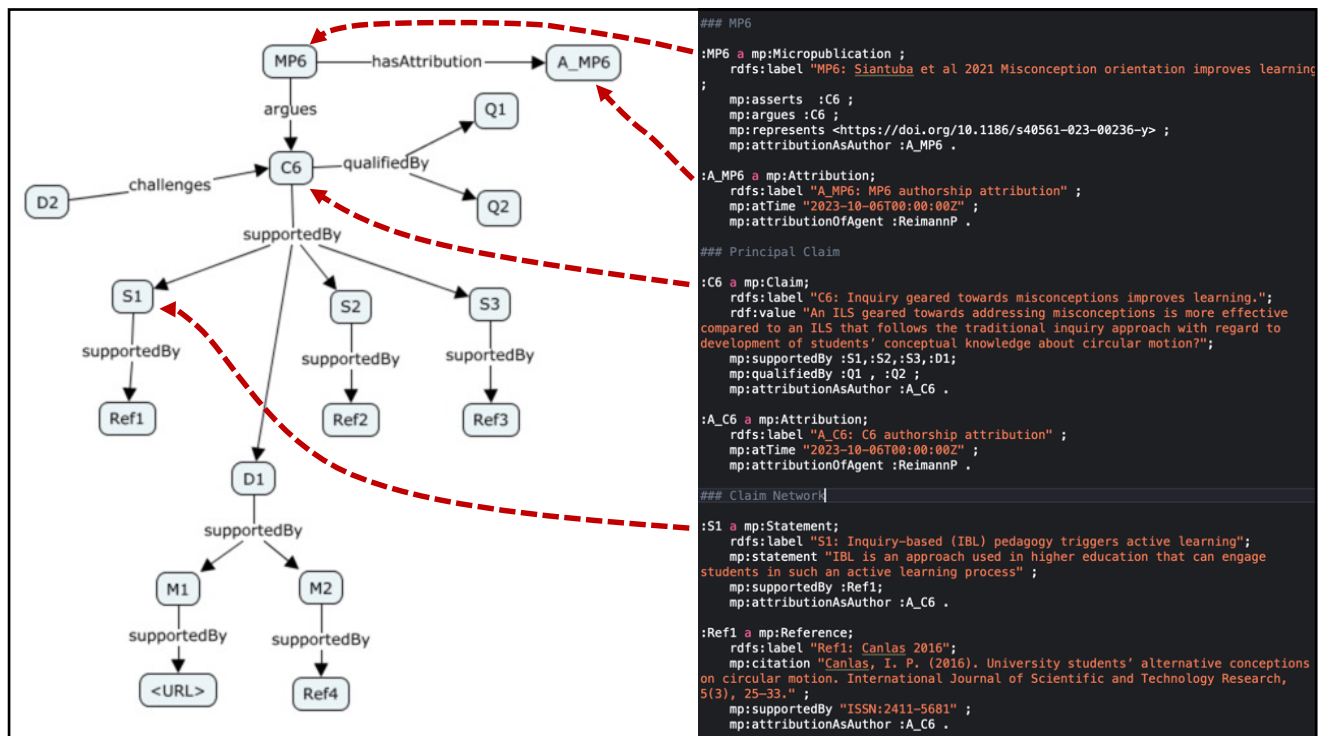
To effectively engage the students in changing their alternative conceptions, it is important to use a learning approach that compels them to actively reconsider their existing conceptions and to construct new conceptions (Ching, 1999; Furtak et al., 2012; Halloun & Hestenes, 1985). IBL is an approach used in higher education that can engage students in such an active learning process (Canlas, 2016; Ching, 1999; Spronken-Smith, 2012). Spronken-Smith (2012) described IBL as a process in which students are actively engaged in knowledge creation that pushes them towards self-directed learning. In IBL, students are compelled to be responsible for their learning by challenging their initial ideas against self-generated data and by actively restructuring their knowledge base (Abd-El-Khalick et al., 2004; de Jong, 2006). In IBL, students often follow an inquiry cycle, bringing them from conceptualization, through experiment design and investigations, to data analysis and reconceptualization (de Jong, 2022; Pedaste et al., 2015). In this way, IBL is also seen as a way to confront students with their naïve conceptions and stimulate them to exchange such conceptions for theoretically sound notions (Kapici et al., 2019; Brod et al., 2018; Maskiewicz & Lineback, 2013; Posner et al., 1982). IBL is traditionally done with real materials and equipment, but nowadays simulations or digital labs are seen as an effective alternative (de Jong, 2022; de Jong et al., 2013).

Claim

Support



59



60

Display all support relations:

```
CONSTRUCT {?a mp:supportedBy ?b}
WHERE {
  ?a mp:supportedBy ?b
}
```

Subject	Predicate	Object
C6	supportedBy	S1
C6	supportedBy	S2
C6	supportedBy	S3
C6	supportedBy	D1
S1	supportedBy	Ref1
Ref1	supportedBy	Canlas2016
S2	supportedBy	Ref2
Ref2	supportedBy	0034654312457206
S3	supportedBy	Ref3
Ref3	supportedBy	s10956-006-9025-3
D1	supportedBy	M1
D1	supportedBy	M2
M1	supportedBy	Sec2
M2	supportedBy	Ref4

61

Find all supporting evidence for a particular claim.

```
prefix xsd: <http://www.w3.org/2001/XMLSchema#>
prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#>
prefix rdfg: <http://www.w3.org/2004/03/trix/rdfg-1/>
prefix dct: <http://purl.org/dc/terms/>
prefix foaf: <http://xmlns.com/foaf/spec/0.98#>
prefix prov: <http://www.w3.org/ns/prov#>
prefix obo: <http://purl.obolibrary.org/obo#>
prefix mp: <http://purl.org/mp/>
prefix : <http://www.learn-web.com/litgraph/>
```

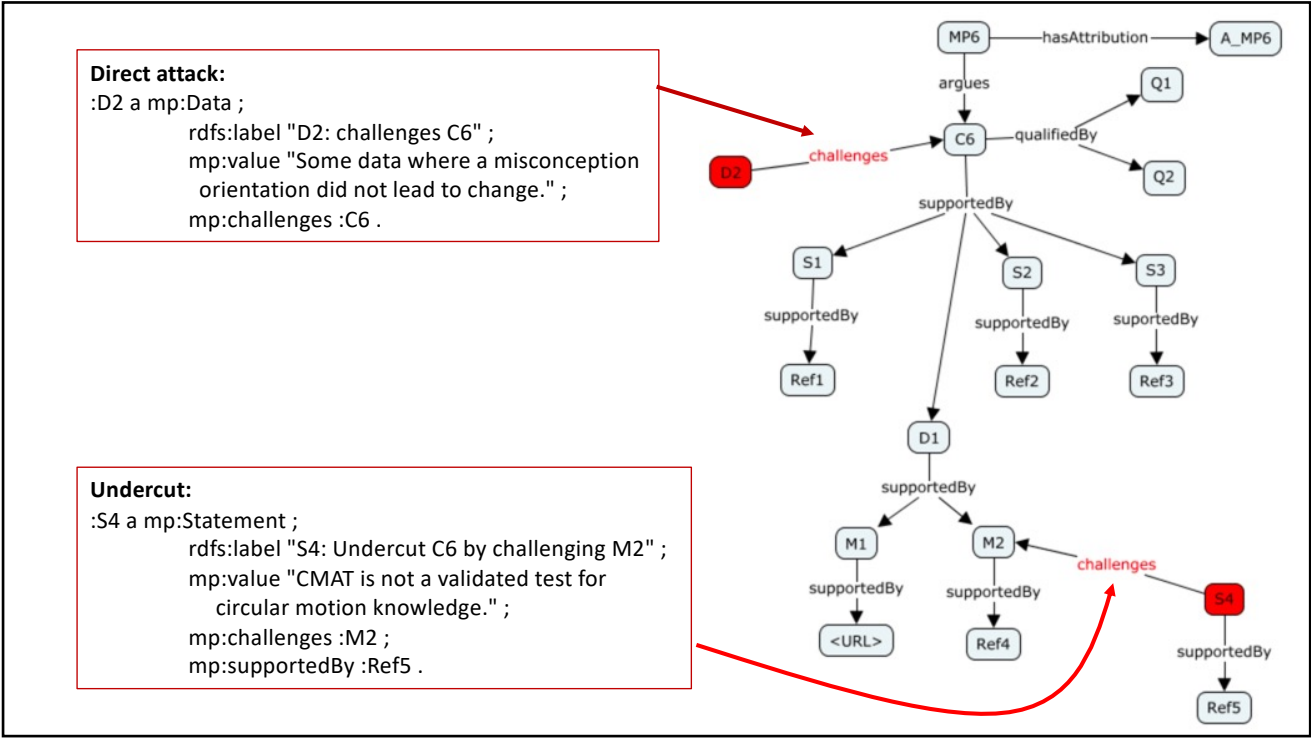
```
SELECT ?a WHERE {
  :C6 mp:supportedBy+ ?a
}
```

Property path
search

a

Ref4
Sec2
s10956-006-9025-3
0034654312457206
Canlas2016
M2
M1
Ref3
Ref2
Ref1
D1
S3
S2
S1

62



63

Direct attack:

```
SELECT ?pro ?con WHERE {  
  {C6 mp:supportedBy ?pro . }  
  UNION {?con mp:challenges :C6 }  
}
```

pro	con
S1	
S2	
S3	
D1	
	D2

Undercut:

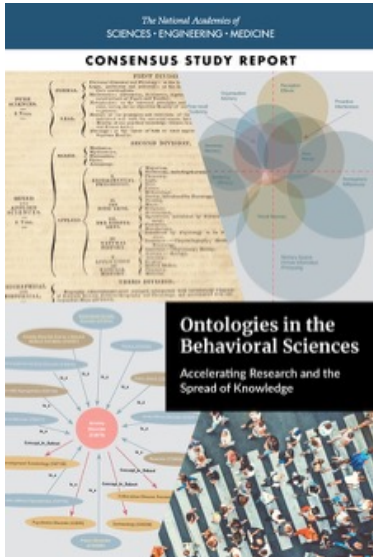
```
SELECT ?claim ?support ?attack WHERE {  
  ?claim mp:supportedBy+ ?support .  
  ?attack mp:challenges ?support . }
```

claim	support	attack
C6	M2	S4
D1	M2	S4

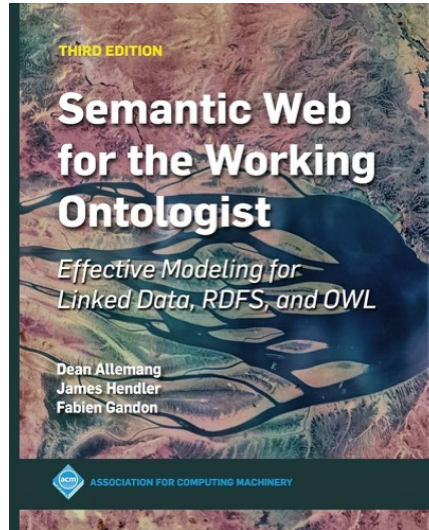
64

Three recommended readings

Why?
(Implementation)



How? (Conceptual)



How?

